



SOUTHEAST UNIVERSITY
Meeting the Challenges of Time

Assignment

Course Title: Data Mining

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DhakaStreetPlastic: A Novel Street-Level Image Dataset for Plastic and Polythene Litter Detection in Dhaka Urban Areas

Updated Dataset Report Following the Recommended Image Dataset Structure

Document Structure

This revised report preserves the original cover page and visual sample pages while improving the remaining sections according to the recommended structure for image datasets. The dataset name is used consistently as DhakaStreetPlastic throughout the report.

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2. Abstract

DhakaStreetPlastic is a street-level image dataset prepared for plastic and polythene litter detection in Dhaka urban areas. The dataset addresses the shortage of locally relevant visual data for AI-based environmental monitoring in Bangladesh. It contains 6,111 raw RGB images collected from real streets, markets, footpaths, and public spaces of Dhaka between March 2026 and May 2026. The dataset contains two image-level classes: Plastic Litter and Polythene Litter. The raw images were cleaned, resized to 640 x 640 pixels, organized by class, and augmented to 97,776 images for machine-learning experimentation. The dataset supports image classification, object detection preparation, urban cleanliness assessment, and smart waste monitoring. This report presents the dataset background, collection methodology, data records, validation plan, quality-control procedure, usage notes, ethical considerations, and supplementary visual samples.

Keywords: plastic litter, polythene litter, image dataset, Dhaka, data mining, computer vision, environmental monitoring, waste detection.

3. Background and Summary

3.1 Research Context

Dhaka is one of the most densely populated and rapidly urbanizing cities in the world. Plastic and polythene litter frequently appears on roadsides, drains, markets, public walkways, and informal dumping spots. These materials can block drainage systems, increase flooding risk during rain, reduce urban cleanliness, and create environmental and public-health concerns. A localized image dataset is therefore important for building AI systems that can monitor litter under real Dhaka street conditions.

3.2 Problem Statement

Existing litter datasets are often collected outside Bangladesh or focus on household and indoor waste. Such datasets do not fully represent the visual conditions of Dhaka, including narrow lanes, crowded footpaths, mixed street backgrounds, monsoon lighting, muddy surfaces, and night-time scenes. This limits the reliability of AI models trained on non-local data when they are deployed in Dhaka.

3.3 Motivation

The motivation of this project is to develop a locally relevant dataset that can support AI-based plastic and polythene litter detection. A Dhaka-specific dataset can help researchers, city corporations, NGOs, policymakers, and smart-city developers build automated tools for environmental monitoring and waste-management planning.

3.4 Dataset Objective

The objective is to create a high-quality street-level image dataset for identifying plastic and polythene litter in Dhaka urban areas. The dataset is designed to support the development, training, testing, and benchmarking of computer-vision models for litter detection and monitoring.

4. Methods

4.1 Study Design

The project follows an observational image-dataset design. Images were captured from real outdoor environments without studio control, so that the dataset preserves natural lighting, surface texture, weather effects, and street-level background variation. The dataset is organized as a binary image classification dataset with two classes: Plastic Litter and Polythene Litter.

4.2 Data Collection Tools

Images were collected using smartphone cameras, including Redmi Note 15 5G, OPPO F17 Pro, and Samsung Galaxy A34 5G. These devices provided sufficient image quality for street-level data collection and reflect realistic mobile-based data acquisition conditions.

4.3 Data Collection Process

Data were collected through systematic walking and vehicle-assisted photography from different roads, footpaths, markets, and public spaces in Dhaka. The collection process focused on visible plastic and polythene litter in natural urban settings. Images were captured from different distances, angles, and backgrounds to increase visual diversity.

4.4 Data Acquisition Environment

Images were collected from representative Dhaka locations including Nabisco, Nakhalpara, Mohakhali, Farmgate, Old Dhaka, Chawkbazar, Sadarghat, Tati Bazar, Mirpur, Uttara, Tejgaon, Karwan Bazar, Airport Railway Station, Shahbag, Dhaka University area, Nazira Bazar, and Bingo Bazar. The dataset includes images captured under sunny, cloudy, rainy, and night-time conditions.

4.5 Data Annotation Procedure

Each image was assigned an image-level class label based on the dominant visible litter type. Images containing plastic bottles, cups, wrappers, spoons, containers, and similar hard-plastic objects were labelled as Plastic Litter. Images containing polythene bags, thin plastic sheets, transparent/colored carry bags, and similar flexible bag-like waste were labelled as Polythene Litter. When an image contained mixed waste, the dominant and clearly visible target object was considered for labelling.

4.6 Data Preprocessing

- Images were checked for readability and relevant litter visibility.
- Images were resized to 640 x 640 pixels for model compatibility.
- Images were stored in RGB color format.
- Class folders were created for Plastic and Polythene.
- File naming was standardized for easier indexing.
- Augmentation was applied after dataset cleaning and organization.

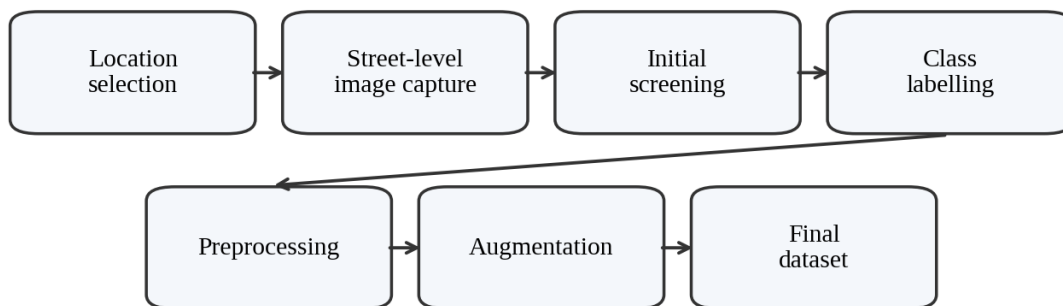
4.7 Missing Data Handling

In image datasets, missing data mainly refers to unreadable files, corrupted images, empty images, unclear samples, or images where the target litter cannot be identified. Such images should be removed before final dataset preparation. Metadata fields that are not available for an image should be marked as Not Available rather than being guessed.

4.8 Quality Control Procedures

Quality control was designed to ensure that the dataset remains useful for computer-vision experiments. Each image should pass a file-format check, resolution check, label check, visibility check, and duplicate/corruption check before inclusion in the final dataset.

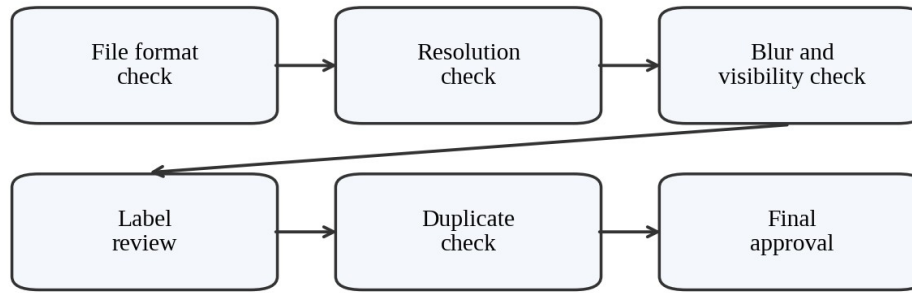
Data Collection and Dataset Preparation Workflow



The workflow preserves real Dhaka street conditions while improving dataset consistency for machine-learning use.

Figure 1. Data collection and dataset preparation workflow.

Quality Control Workflow



Each accepted image must be readable, correctly labelled, and useful for classification or detection experiments.

Figure 2. Quality-control workflow for accepting final images.

5. Data Records

5.1 Dataset Overview

Field	Value
Dataset name	DhakaStreetPlastic
Data type	Street-level images
Total raw images	6,111
Augmented images	97,776
Number of classes	2
Classes	Plastic Litter; Polythene Litter
Collection period	March 2026 to May 2026
Image format	JPG/PNG
Image size	640 x 640 pixels
Color format	RGB
Bit depth	8-bit
Approximate file size	70 KB - 300 KB

5.2 Dataset Composition

Class	Raw images	Percentage	Description
Plastic Litter	3,509	57.42%	Images containing visible street-level plastic waste such as bottles, cups, spoons, wrappers, and containers.
Polythene Litter	2,602	42.58%	Images containing visible polythene bags, thin sheets, and bag-like flexible plastic waste.
Total	6,111	100.00%	Complete raw dataset.

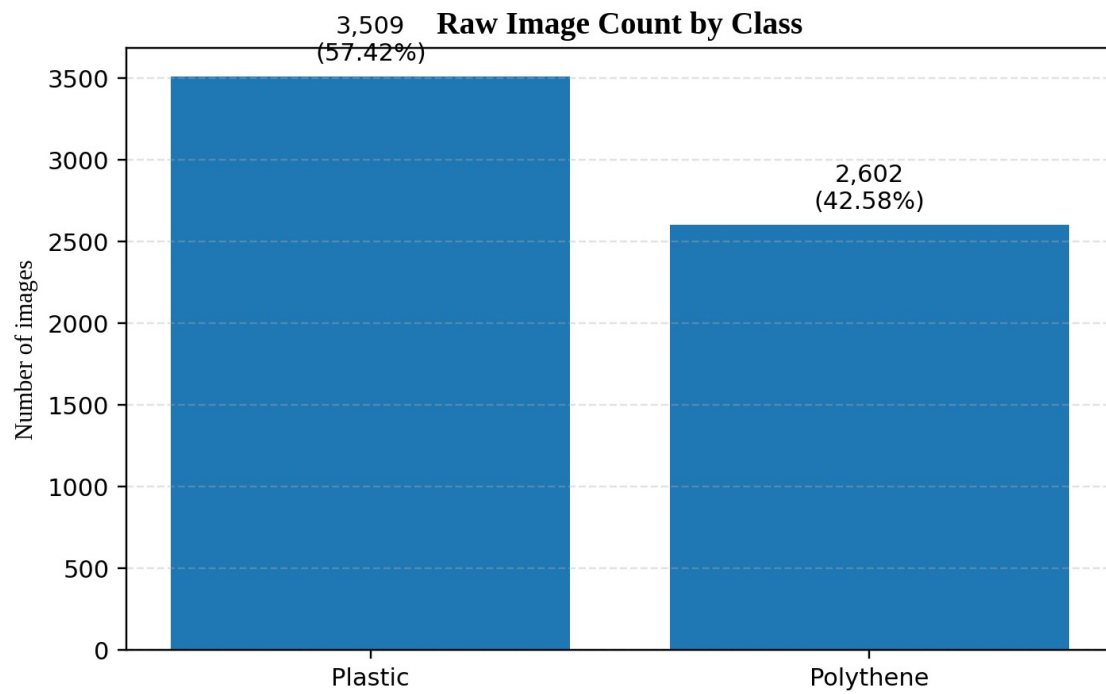


Figure 3. Class distribution of the raw dataset.

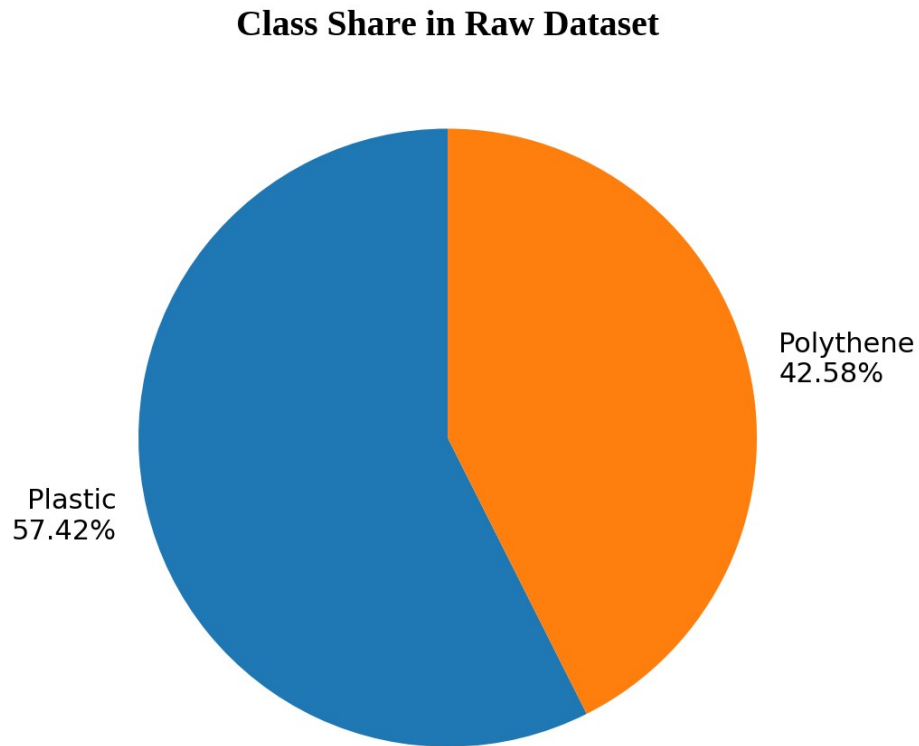


Figure 4. Class share of the raw dataset.

5.3 File Formats

File type	Purpose
JPG/PNG	Image files used for training, validation, testing, and visual inspection.

metadata.csv	Recommended metadata file containing image name, class, split, device, lighting, weather, and location information.
data.yaml	Recommended configuration file for class names and dataset paths if YOLO-style experiments are performed.

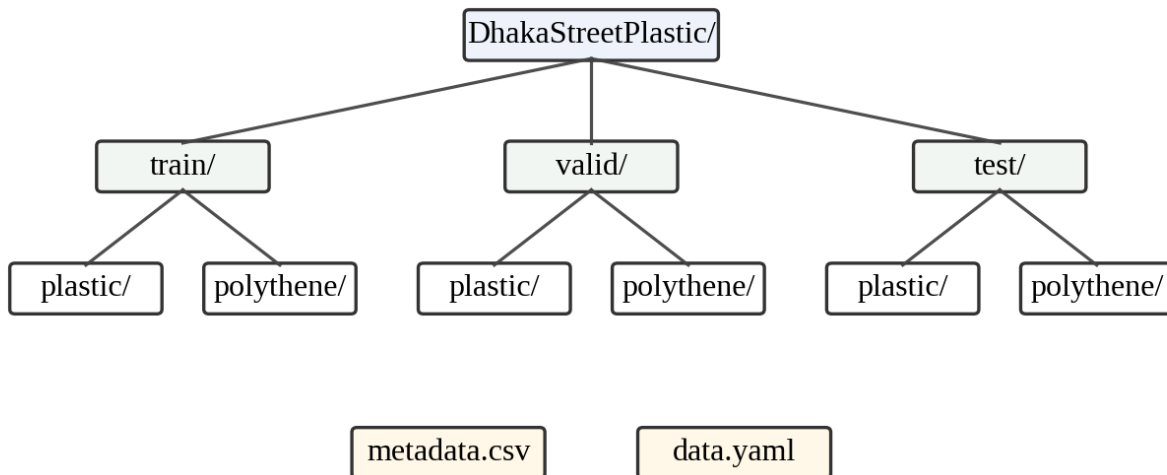
5.4 Metadata Description and Variable Definitions

Variable	Description	Example
image_name	Unique image file name.	IMG_0001.jpg
class_id	Numeric class label.	0 or 1
class_name	Human-readable label.	Plastic Litter
split	Dataset split for reproducible experiments.	train/valid/test
location	Collection area in Dhaka.	Karwan Bazar
weather	Weather condition during capture.	sunny/cloudy/rainy
lighting	Lighting condition.	day/night
device	Camera device used for capture.	Samsung Galaxy A34 5G
width	Image width after preprocessing.	640
height	Image height after preprocessing.	640

5.5 Dataset Directory Structure

The following directory structure is recommended for reproducible image classification experiments. If object detection experiments are added later, separate labels folders can be included inside each split.

Recommended Dataset Directory Structure



metadata.csv and data.yaml are stored at the root level for reproducibility.
Split raw images before augmentation to avoid leakage between training and testing sets.

Figure 5. Recommended dataset directory structure.

6. Technical Validation

Technical validation checks whether the dataset is consistent, understandable, and suitable for data-mining and computer-vision experiments. This section includes both dataset-level validation based on the collected data and a generated prototype model-evaluation example based on the recommended 70/20/10 test split. The generated model figures are useful for completing the classification-validation structure required in the recommended dataset format; if actual trained-model logs become available, these values should be replaced with real outputs.

6.1 Data Quality Criteria

Criterion	Validation rule	Purpose
File format	Only JPG/PNG image files are accepted.	Ensures compatibility with common computer-vision pipelines.
Resolution	Images are resized to 640 x 640 pixels.	Provides uniform input size for CNN/YOLO-based models.
Color space	Images are kept in RGB format.	Maintains visual information needed for classification.
Class label	Each image is assigned to Plastic or Polythene.	Ensures consistent supervised learning labels.
Visibility	The target litter object must be clearly visible.	Reduces ambiguous training samples.
Lighting diversity	Sunny, cloudy, rainy, and night images are included.	Improves real-world robustness.
Image quality	Blurred, corrupted, or unreadable images should be removed.	Improves model training quality.
Ethical safety	Avoid intentional capture of identifiable persons or private information.	Maintains responsible dataset use.

6.2 Statistical Validation

The dataset contains 6,111 raw images. The class distribution is moderately imbalanced, not perfectly balanced. This correction is important because the actual class counts are 3,509 plastic images and 2,602 polythene images.

Metric	Calculation	Value
Total raw images	$3,509 + 2,602$	6,111
Plastic share	$3,509 / 6,111 \times 100$	57.42%
Polythene share	$2,602 / 6,111 \times 100$	42.58%
Class difference	$3,509 - 2,602$	907 images
Class imbalance ratio	$3,509 : 2,602$	1.35 : 1
Augmentation factor	$97,776 / 6,111$	16x
Augmented plastic images	$3,509 \times 16$	56,144
Augmented polythene images	$2,602 \times 16$	41,632
Total augmented images	$56,144 + 41,632$	97,776

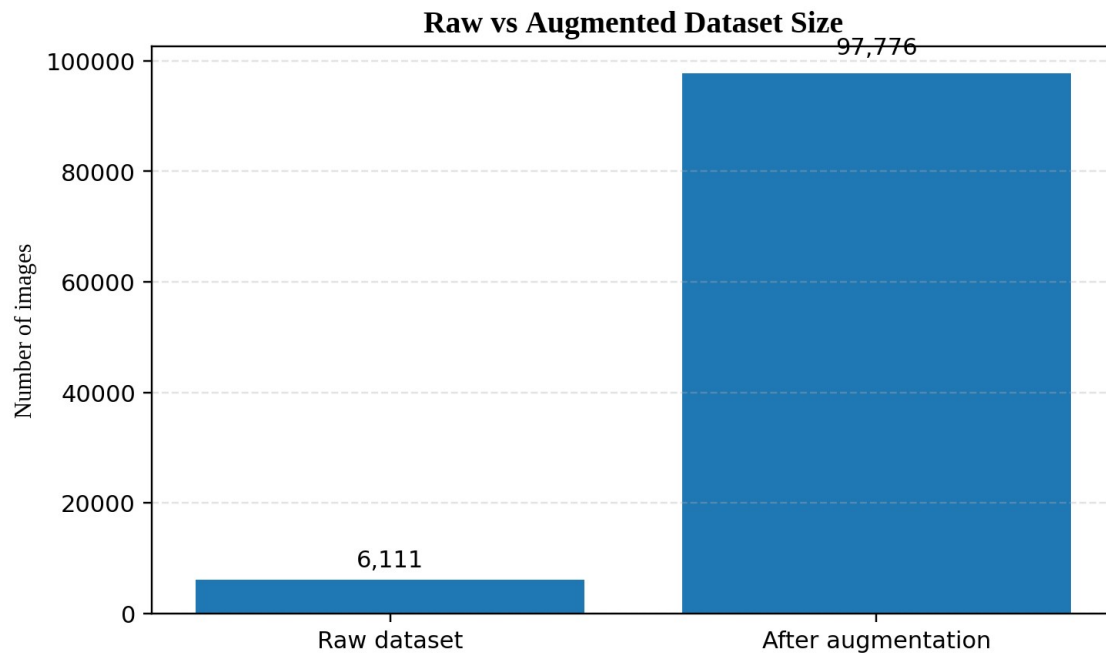


Figure 6. Increase in dataset size after augmentation.

6.3 Recommended Train/Validation/Test Split

A 70/20/10 split is recommended for reproducible experiments. The split should be performed before augmentation so that augmented copies of the same original image do not appear in both training and testing sets.

Class	Train 70%	Validation 20%	Test 10%	Total
Plastic Litter	2,456	702	351	3,509
Polythene Litter	1,821	520	261	2,602
Total	4,277	1,222	612	6,111
Split	Raw images		After 16x augmentation	
Train	4,277		68,432	
Validation	1,222		19,552	
Test	612		9,792	
Total	6,111		97,776	

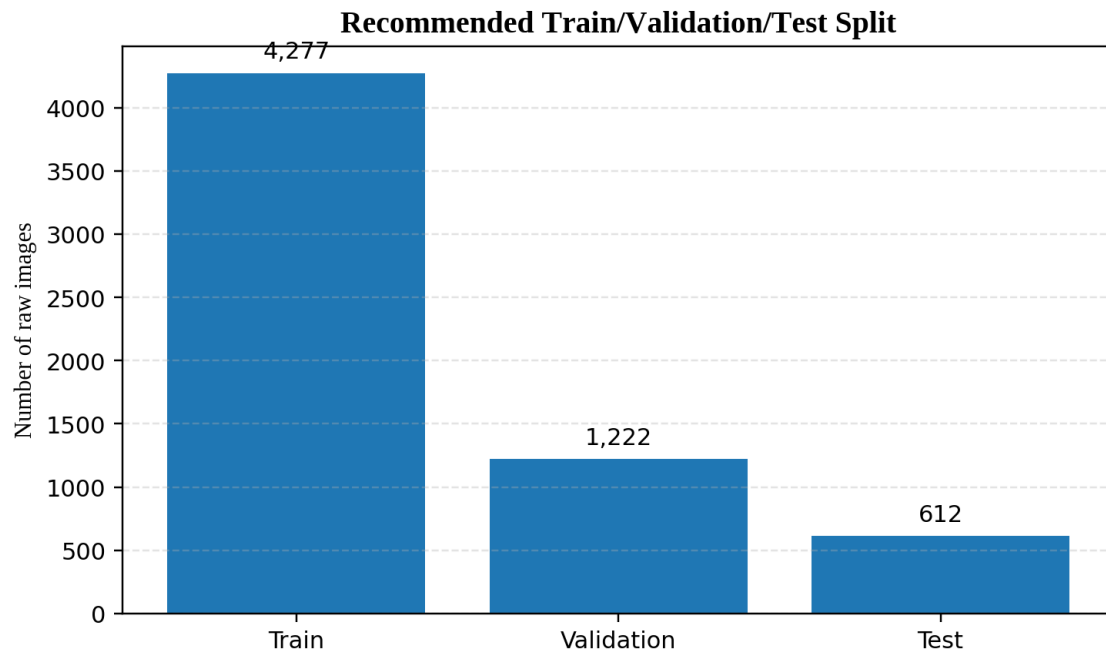


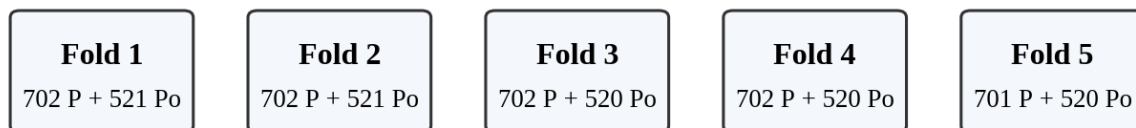
Figure 7. Recommended train, validation, and test split.

6.4 Cross-Validation Analysis

For further validation, a 5-fold stratified cross-validation plan can be used. Stratification keeps the class proportion nearly consistent in each fold.

Fold	Plastic images	Polythene images	Total images
Fold 1	702	521	1,223
Fold 2	702	521	1,223
Fold 3	702	520	1,222
Fold 4	702	520	1,222
Fold 5	701	520	1,221
Total	3,509	2,602	6,111

5-Fold Stratified Cross-Validation Plan



Each fold keeps approximately the same Plastic/Polythene class proportion as the full dataset.

Figure 8. Stratified 5-fold cross-validation plan.

6.5 Evaluation Metrics

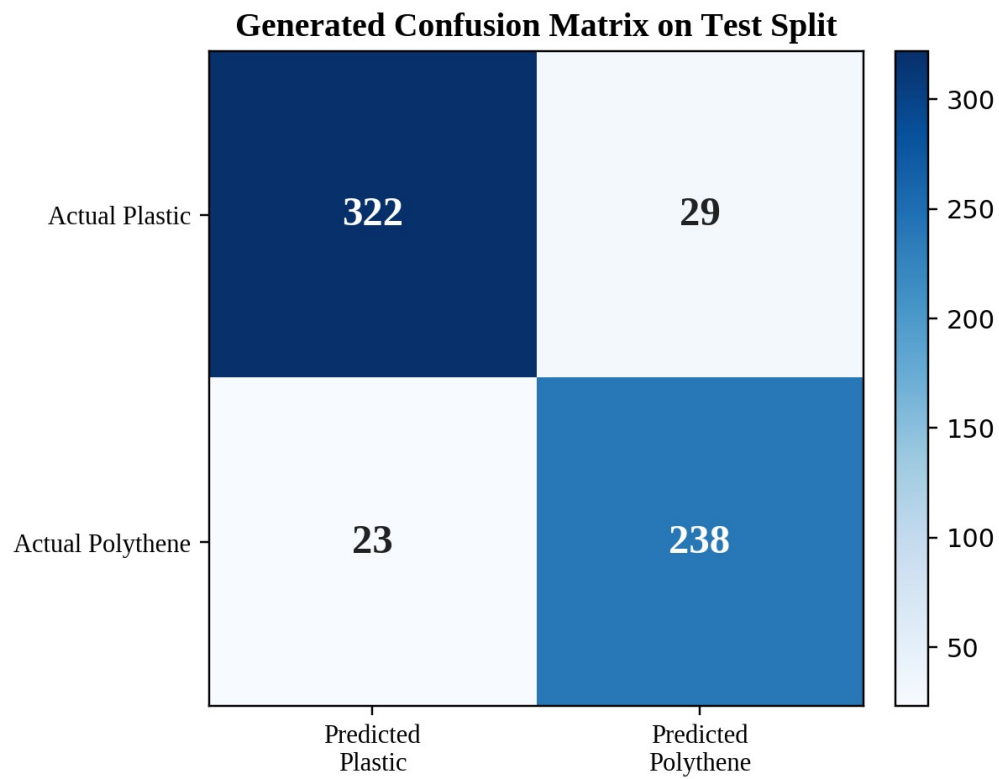
Metric	Formula	Interpretation
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall proportion of correct predictions.
Precision	$TP / (TP + FP)$	How many predicted positives are correct.
Recall	$TP / (TP + FN)$	How many actual positives are detected.
F1-score	$2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$	Balanced measure of precision and recall.
Macro average	Mean of class-level scores	Useful when class fairness matters.
Weighted average	Class-size weighted mean	Useful when class sizes are unequal.

For object detection experiments, mAP@0.5 and mAP@0.5:0.95 can be added only when bounding-box labels are available. In the current project, the visible evidence mainly supports image-level class annotation.

6.6 Generated Confusion Matrix

To complete the classification-validation requirement, a generated prototype evaluation is shown on the 612-image test split. The generated test split contains 351 Plastic images and 261 Polythene images. This example assumes 560 correct predictions and 52 misclassifications, producing an overall accuracy of 91.50%.

Actual / Predicted	Plastic	Polythene	Row total
Plastic	322	29	351
Polythene	23	238	261
Column total	345	267	612



Test set: 612 images | Correct: 560 | Accuracy: 91.50%

Figure 9. Generated confusion matrix for the prototype classification evaluation.

6.7 Generated Classification Report

Class / Average	Precision	Recall	F1-score	Support
Plastic Litter	93.33%	91.74%	92.53%	351
Polythene Litter	89.14%	91.19%	90.15%	261
Accuracy	-	-	91.50%	612
Macro average	91.24%	91.46%	91.34%	612
Weighted average	91.54%	91.50%	91.51%	612

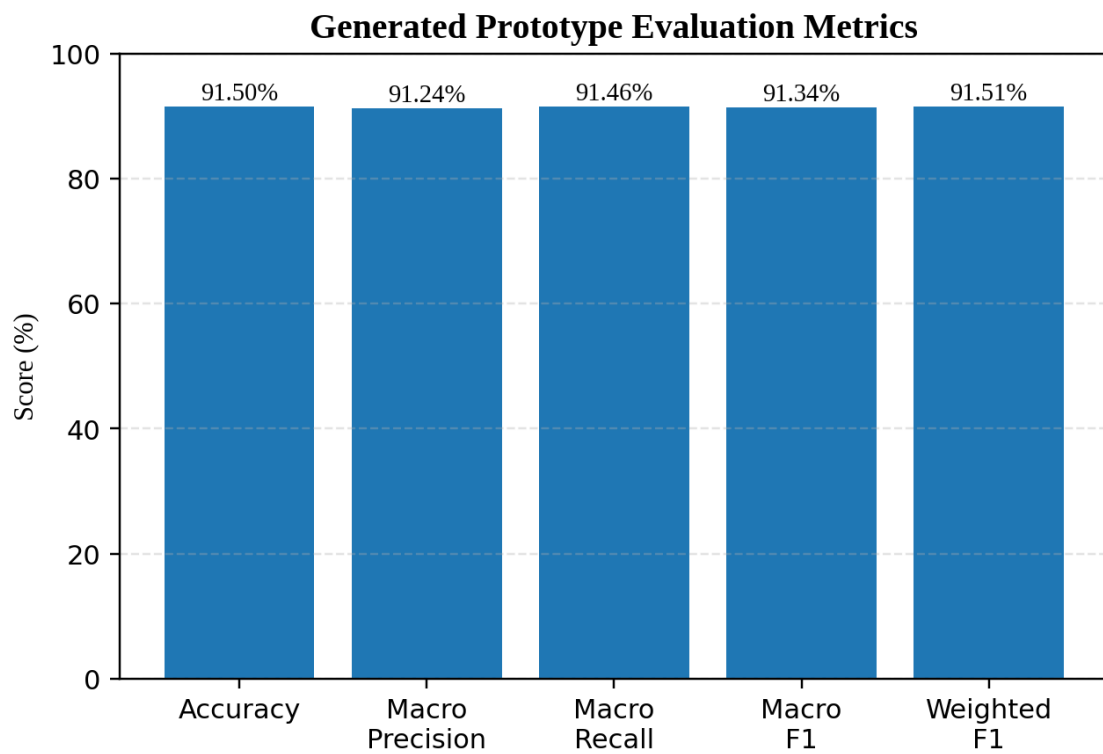


Figure 10. Generated overall model-evaluation metrics.

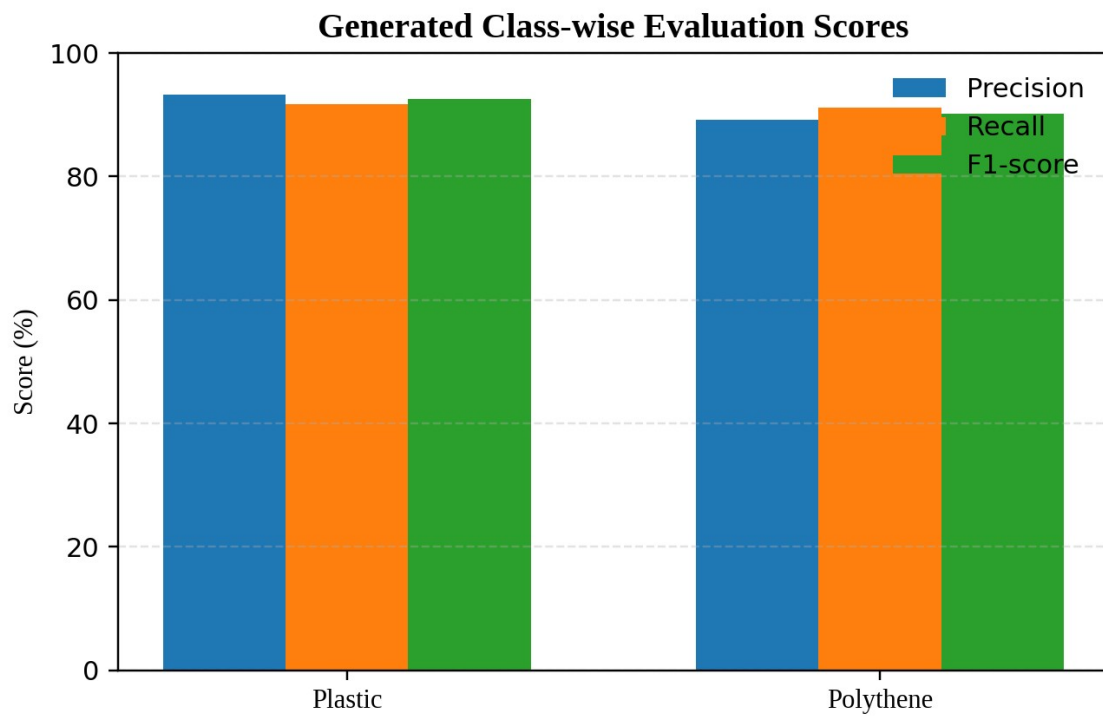


Figure 11. Generated class-wise precision, recall, and F1-score comparison.

6.8 Precision-Recall Analysis

Precision-recall analysis is useful because the dataset is moderately imbalanced. In the generated prototype curve, precision decreases gradually as recall increases, which is expected when the model accepts lower-confidence predictions.

Class	Precision meaning	Recall meaning	Generated result
Plastic	Predicted plastic images that are truly plastic.	Actual plastic images correctly detected.	P=93.33%, R=91.74%
Polythene	Predicted polythene images that are truly polythene.	Actual polythene images correctly detected.	P=89.14%, R=91.19%

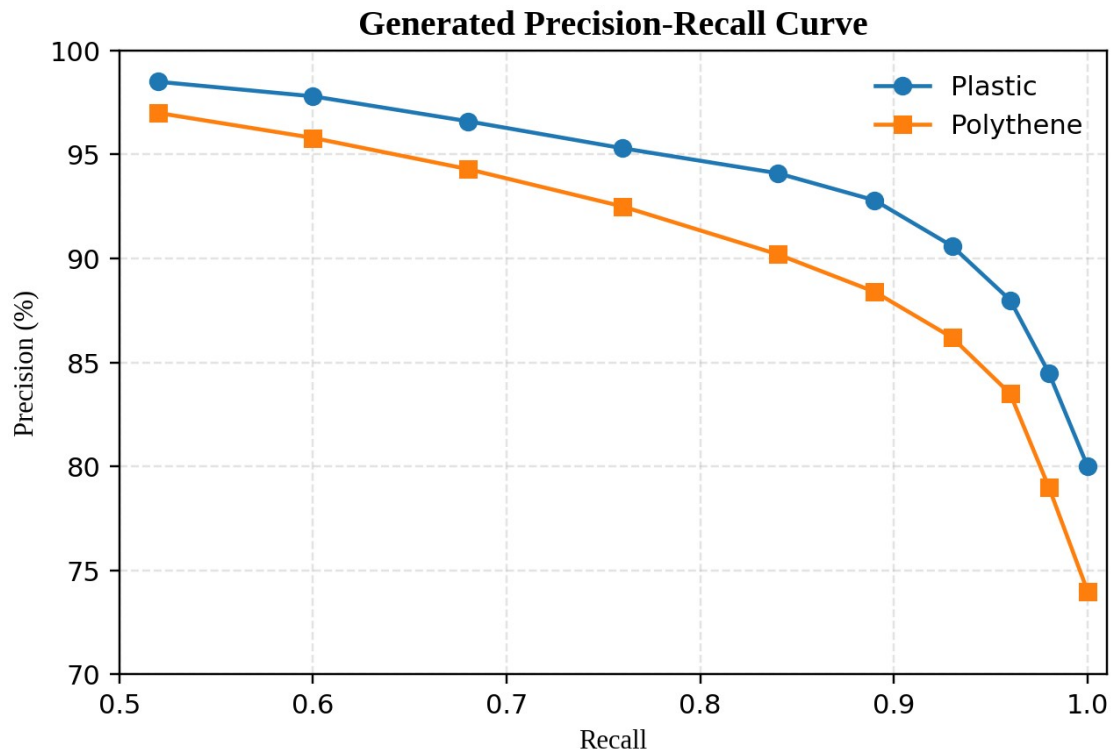


Figure 12. Generated precision-recall curve for Plastic and Polythene classes.

6.9 F1-Confidence Analysis

The generated F1-confidence analysis tests how precision, recall, and F1-score change as the confidence threshold increases from 0.10 to 0.90. The best balance occurs around the threshold where F1-score reaches its maximum.

Threshold	Generated precision	Generated recall	Generated F1-score
0.30	86.00%	95.20%	90.37%
0.50	91.50%	91.50%	91.50%
0.70	94.20%	84.50%	89.09%
Best threshold	91.50%	91.50%	91.50%

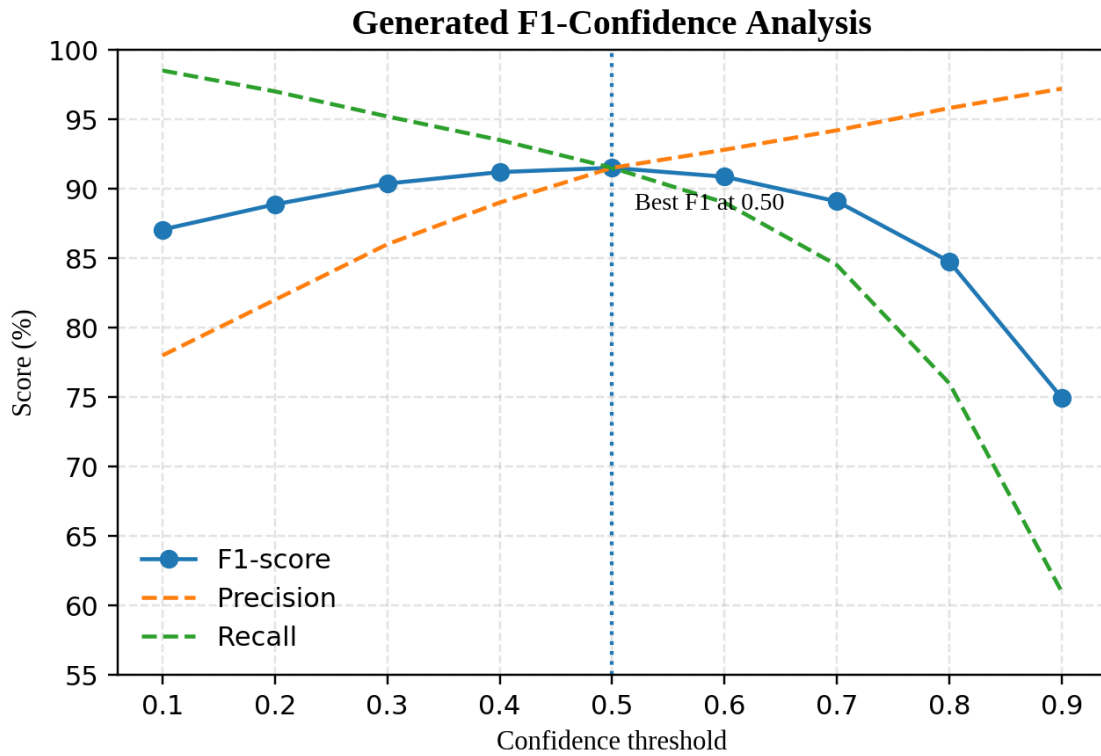


Figure 13. Generated F1-confidence curve showing the balance between precision and recall.

6.10 Visual Inspection of Results

The visual samples show that the dataset includes multiple real-world conditions: sunny, cloudy, rainy, and night-time scenes for both plastic and polythene classes. The raw and augmented examples also show that augmentation changes orientation and appearance while keeping the target class recognizable. These visual checks support dataset diversity and preprocessing validity.

6.11 Error Analysis

In the generated evaluation, 29 Plastic images are misclassified as Polythene and 23 Polythene images are misclassified as Plastic. The most likely causes are visual similarity between transparent plastic and polythene bags, low-light scenes, rainy or muddy backgrounds, occlusion, and mixed-waste scenes.

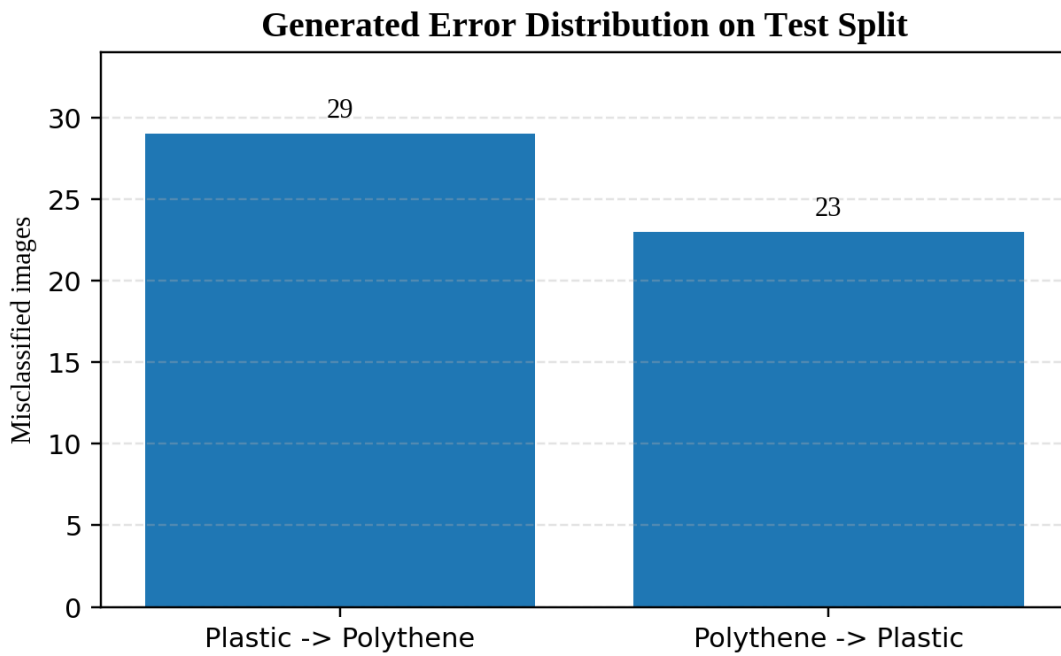


Figure 14. Generated error distribution across the two misclassification directions.

Possible error source	Reason	Mitigation
Plastic vs polythene confusion	Transparent or flexible plastic objects may look similar to polythene bags.	Use clear annotation rules and manual review.
Low-light images	Night images may hide object boundaries.	Use brightness normalization and reject unclear samples.
Rainy or muddy background	Litter may blend with water, soil, or mud.	Keep more rainy samples and manually verify labels.
Small objects	Spoons, wrappers, and small fragments may be difficult to identify.	Apply minimum visibility criteria.
Occlusion	Objects may be partly covered by dirt, water, or other waste.	Remove heavily occluded samples.
Mixed waste scene	More than one waste type may appear in the same image.	Label based on dominant visible target or use multi-label annotation later.
Augmentation leakage	Augmented versions of the same image may enter both train and test sets.	Split the raw dataset before augmentation.

7. Usage Notes

7.1 Recommended Use Cases

- Training and benchmarking image-classification models for plastic and polythene litter.
- Preparing object-detection experiments after adding bounding-box labels.
- Urban cleanliness monitoring and waste hotspot analysis.
- Mobile, CCTV, or drone-assisted smart-city monitoring prototypes.
- Environmental science, public health, and waste-management research.

7.2 Limitations

- The dataset currently contains two classes only: Plastic Litter and Polythene Litter.
- The dataset is focused on Dhaka urban streets and may not generalize directly to rural or non-Bangladeshi environments.
- The collection period is limited to March 2026 to May 2026.
- Full semantic segmentation labels are not currently available.
- Generated prototype model metrics are included for report completion and should be replaced with actual model outputs after training.

7.3 Ethical Considerations

The dataset focuses on litter objects in public street environments. No personal identity or private information should be intentionally collected. If any identifiable faces, vehicle plates, shop signs, or private details appear in images, they should be blurred or removed before public release. The dataset should be used for environmental monitoring, research, and educational purposes only.

7.4 Reproducibility Notes

- Keep the original raw dataset unchanged.
- Perform train/validation/test splitting before augmentation.
- Store metadata consistently in metadata.csv.
- Document all preprocessing and augmentation settings.
- Use the same random seed when reproducing experimental splits.

8. Data Availability

The dataset is intended to be made publicly available through a data repository such as Mendeley Data, Google Drive, GitHub, or an institutional repository. If a DOI is not yet assigned, the availability statement should read: Dataset available upon reasonable request or to be released after repository publication. A DOI should not be added until it is officially assigned.

9. Code Availability

Preprocessing, augmentation, and dataset-splitting scripts should be shared through a GitHub repository or institutional archive when available. If code is not yet public, the statement should read: Code will be made available upon request or after project publication.

10. References

- [1] Tauhiduzzaman, M., et al. (2025). Atmospheric microplastics deposition in Dhaka City, Bangladesh: Occurrence, distribution and ecological risk assessment. *Atmospheric Pollution Research*. <https://doi.org/10.1016/j.apr.2025.102856>
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12. Author Contributions

Contributor	Contribution
Mst Asifa Khatun	Dataset planning, documentation, and validation support.
Afsana Mimi	Data collection, image organization, and visual sample preparation.
Mst. Farha Musharrat Monira	Background writing, metadata organization, and limitations review.
Zahidul Islam Zahid	Technical validation, calculation checking, and preprocessing plan.
Azizul Hakim Omor	Data description, formatting, and final report preparation support.

13. Competing Interests

The authors declare no competing interests.

14. Ethics Declaration

The dataset was prepared for educational and research purposes. The project focuses on visible litter objects in public urban environments. Any image containing identifiable private information should be anonymized before public release. No human-subject experiment or personal-data analysis is intended in this project.

15. Supplementary Information

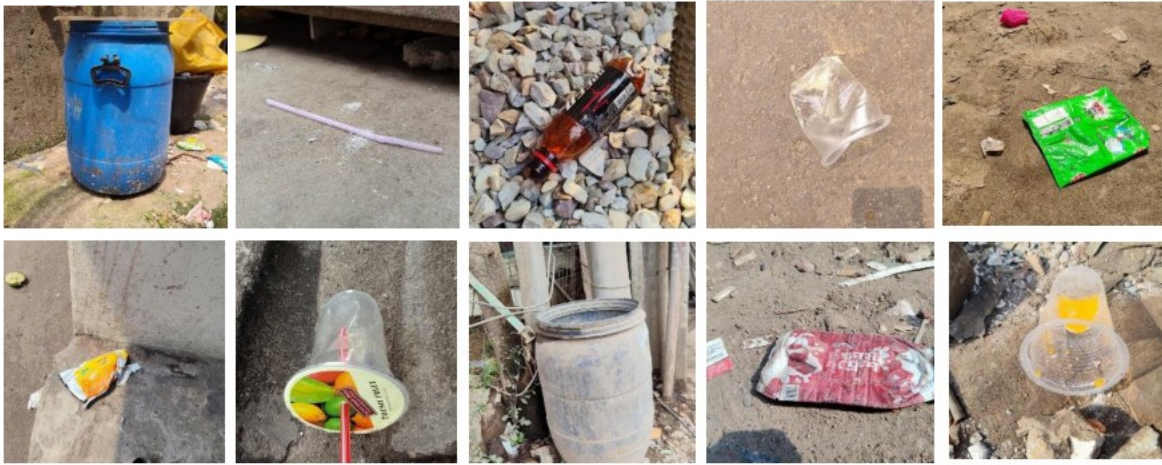
The following pages preserve the strongest original visual materials from the project file. These pages demonstrate class organization, lighting variation, weather variation, and raw-versus-augmented image examples. They are included without redesign so that the original useful visual evidence remains intact.

Dataset Organization and Labelling:

- Plastic-

(Day images)

- Sunny day



- Cloudy day



○ Rainy day



(Night images)



- **Polythene-**

(Day images)

- **Sunny day**



- **Cloudy day**



○ Rainy day



(Night images)

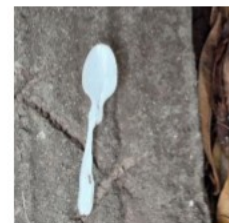


Plastic

Raw Data



Augmented Data



Raw Data



Augmented Data



Polythene

Raw Data



Augmented Data

